PYTHON PROJECT

EVALUATION – 1 REPORT

ON

CYBER LEGAL GUIDANCE SYSTEM

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Chapter 1: ABSTRACT

Problem Statement:
Due to legal illiteracy and complicated procedural requirements, victims of cyber fraud frequently miss the crucial "Golden Hour" for fund recovery as the practice grows more common in India's digital economy. Users are unable to recognise accurate citations under the recently introduced Bharatiya Nyaya Sanhita (BNS) because current AI systems are unable to comprehend informal, code-mixed Hinglish queries. Localised, automated tools that offer prompt, customised legal advice and pre-filled procedural handouts for digital fraud scenarios are severely lacking.

Analysis:
Although RAG (Retrieval-Augmented Generation) and Legal-BERT models work well for standardised English, they encounter serious accuracy issues with Indian code-mixed dialects and the 2024 switch from the IPC to the BNS, according to a thorough review of the literature. The majority of legal chatbots function as general conversationalists and lack the specialised forensic ability to extract actionable evidence from unstructured user conversations, including wallet addresses, transaction timestamps, and UTR numbers.

Proposed System:
The Cyber Legal Guidance System is a system that functions as a specialised digital information handout generator.

Chapter 2: INTRODUCTION

The rapid digitization of the Indian economy has brought millions of first-time internet users into the digital fold, unfortunately making them prime targets for sophisticated cyber-fraud syndicates. Despite the scale of the threat, a massive "Legal Literacy Gap" exists. The recent overhaul of India’s criminal justice system, replacing the colonial-era Indian Penal Code (IPC) with the **Bharatiya Nyaya Sanhita (BNS)** effective July 2024, has introduced new challenges for the common citizen. Victims often find it difficult to articulate their grievances in formal legal English, instead using "Hinglish", a blend of Hindi and English.

It is suggested that the Cyber Legal Guidance System serve as a link between the victim's informal story and the official legal system. This system is a domain-specific "Legal First-Aid" tool, in contrast to generic AI. It is intended to reduce the amount of time that passes between a crime happening and being reported. The system offers an organised recovery path by using a Retrieval-Augmented Generation (RAG) framework to access verified legal databases and Natural Language Processing (NLP) to parse mixed-script queries. It helps the user comprehend the possible course of their legal battle by not only identifying the pertinent sections of the BNS but also contextualising them using public records of court cases.

Chapter 3: LITERATURE REVIEW

The following table summarizes the key research and frameworks relevant to this project:

S.No	Year & Author	Title of the Paper	Methodology Used	Dataset Used	Results Achieved
I.	Legal AI & System Architecture				
1	2025, B. Patil et al.	BNS Mitra: RAG-optimized LLM based AI-powered Legal Virtual Assistant	RAG framework using FAISS for vector storage and LLMs	Bharatiya Nyaya Sanhita (BNS) documents	Provides context-aware assistance for law enforcement and the public
2	2025, D. Panchal et al.	LawPal: A Retrieval Augmented Generation Based System for Enhanced Legal Accessibility in India	RAG-based chatbot using FAISS vectorstore	Indian Constitution, legal books, and official documentation	Efficient and accurate legal information retrieval
3	2025, A. M. S. & K. S. Easwararkumar	JudgmentGraph: Benchmarking Legal Knowledge Graph Construction from Supreme Court Rulings	Ontology-guided pipeline using RoBERTa-large with DHA and CRF	Indian Supreme Court judgments; manual gold standard (15 documents)	Precision: 81.4%, Recall: 88.4%, F1-score: 84.7%
4	2025, P. Kumar	A Legal-BERT Validated RAG Framework for Trustworthy AI-Assisted Legal Reasoning	Five-component architecture: LayoutLMv3, BART, DPR, GPT-4, and Legal-BERT validation	Curated legal documents and statutory texts	Hybrid Verification Loop iteratively refines outputs for compliance
5	2025, A. Garlapati et al.	Enhancing Public Access to Legal Knowledge in India: A Legal Chatbot Using Legal BERT, GPT-2, and RAG	Fine-tuning Legal BERT for terminology and GPT-2 for RAG-based responses	Indian Constitution and curated legal corpus	Delivers accurate, contextually appropriate responses to queries
6	2025, M. R. Lima et al.	Question Answering Techniques for Portuguese Legal Documents: A Systematic Literature Review	Systematic Literature Review (SLR) using PRISMA 2020 guidelines	10 primary studies from 9 databases (ACM, Scopus, etc.)	Identified that 70% of relevant research emerged between 2023-2025
7	2024, R. Yang	CASEGPT: A Case Reasoning Framework Based on Language Models and Retrieval-Augmented Generation	Synergizes LLMs and RAG for semantic searches and contextual understanding	Medical and legal field datasets (legal precedents)	12% improvement in precision for legal precedent retrieval
8	2024, Anonymous	AALAWYER: A Generative Retrieval-Augmented LLM System for Legal Reasoning	Generative RAG retrieving legal articles and precedent cases	Large-scale legal corpus and court hearing transcripts	Significant reduction in hallucinated citations
9	2025, S. Ni et al.	Pre-training, Fine-tuning and Re-ranking: A Three-Stage Framework for Legal QA	PFR-LQA framework: Domain pre-training, dual-encoder fine-tuning, and re-ranking	Large-scale database of legal question-answer pairs	Boosted dense retrieval performance and text representation
10	2025, V. Van Long et al.	Combining Large Language Model and Supportive Models for Legal Question Answering	Ensemble: semantic search with BGE-M3 and traditional ranking (BM25, TF-IDF)	ALQAC 2025 competition dataset	Improved retrieval accuracy through LLM-augmented data

II.	**Linguistic Processing (Hinglish & Multilingual NLP)**				
11	2025, R. Sheth et al.	COMI-LINGUA: Expert Annotated Large-Scale Dataset for Multitask NLP in Hindi-English Code-Mixing	Multitask NLP using fine-tuned open-weight LLMs (Llama-3, Qwen)	125K+ annotated instances across five core NLP tasks	Achieved up to 95.25 F1 in NER and 98.77 F1 in MLI
12	2024, A. S. Kapse et al.	IndiTranslate: Bridging Language Barriers in India	Real-time translation using advanced NLP and Transformer models (BERT/GPT)	Linguistic data from government archives and news sites	Achieved over 85% accuracy in real-time translation
13	2024, S. Chakraborty et al.	LINGUABRIDGE: AI-Powered Multilingual Translation & Sentiment Analysis	Modular framework using Google Translate, TextBlob, Llama, and feedback	Multilingual data covering 22 Indian languages	Real-time translation and emotion detection across Indian regions
14	2024, P. Tewari et al.	Hinglish Text Analysis: Challenges and Opportunities in Multilingual NLP	Specialized NLP pipeline: script standardization, SVM, Naive Bayes, LSTM, and BERT	Social media data containing mixed Roman/Devanagari scripts	Outperforms existing methods in classifying unpredictable Hinglish
15	2020, K. Singh et al.	Language Identification and NER in Hinglish Code Mixed Tweets	Token-level language identification and custom NER for code-mixed text	Hinglish (Hindi-English) Twitter data	Outperformed off-the-shelf tools by 33.18% in F1 score
16	2020, N. Garg & K. Sharma	Annotated corpus creation for sentiment analysis in code-mixed Hinglish	Corpus creation with three-point sentiment labeling	Hinglish social network data (YouTube, Facebook)	Established a foundational dataset for Hinglish sentiment
17	2024, A. Wadhawan & A. Aggarwal	Towards Emotion Recognition in Hindi-English Code-Mixed Data	Comparison of CNN, LSTM, Bi-LSTM, and Transformers (BERT, RoBERTa)	Hinglish dataset labeled for emotion detection	BERT outperformed other models with 71.43% accuracy
18	2025, A. Mangla et al.	Code-Mixing and Code-Switching on Social Media Text: A Brief Survey	Survey of NLP for transliteration, Romanization, and language fusion	Social media platforms (Facebook, Google Reviews)	Analyzed linguistic versatility in the Indian landscape
19	2025, A. Mahaganapathy	A survey and evaluation of TTS systems for the Tamil language	Perceptual evaluation framework measuring fatigue and quality using CMOS	Existing Tamil TTS systems and custom perceptual data	Established a benchmark for Tamil voice interfaces
III.	**Cybercrime & Digital Forensics**				
20	2025, M. M. Khazem et al.	Real-Time Detection of Cybercrime and Digital Forensics Automation	Transformer-enhanced CNN-LSTM with blockchain preservation	NSL-KDD, WSN-DS, and IoT-specific datasets	Robust real-time detection and tamper-proof evidence collection
21	2021, J. Adkins et al.	Identifying Persons of Interest in Digital Forensics Using NLP-Based AI	Ensemble of topic modeling, pairwise correlation, and sentiment analysis	Unstructured user-generated data (emails, chats)	Efficiently mined patterns to identify POIs manually missed

22	2021, D. Sun et al.	NLP-based digital forensic investigation platform for online communications	Multiphase platform: vectorization, feature selection, and classifier generation	Email and social media communications for litigation	Efficiently identified suspicious patterns in digital text
23	2019, F. Amato et al.	Analyse digital forensic evidences through a semantic-based methodology	Semantic-based methodology for correlation and reasoning	Computer Forensics domain textual data and logs	Successfully extracted evidence for telematic crime reconstruction
24	2023, R. Debbarma	The Legal Framework and Challenges in Prosecuting Cybercrimes	Qualitative legal research examining hacking, identity theft, and fraud	Case law and statutory frameworks	Identified jurisdictional hurdles and need for tech expertise
IV.	**Legal Frameworks & Victim-Centric Justice**				
25	2024, D. Rajamanickam	Improving Legal Entity Recognition Using a Hybrid Transformer Model	Hybrid model combining Legal-BERT with semantic similarity-based filtering	15,000 annotated legal documents (contracts, court rulings)	State-of-the-art F1 score of 93.4%
26	2025, R. Gulati	Bharatiya Nyaya Sanhita 2023: Relevance and Challenges	Comparative analysis between IPC 1860 and BNS 2023	The BNS, BNSS, and BSA legislative frameworks	Highlighted new cybercrime categories and digital evidence rules
27	2020, S. Gour	Victim-Centric Justice in Criminal Law: A Critical Analysis	Qualitative examination of legal foundations and restorative mechanisms	Indian criminal justice framework and judicial precedents	Defined the shift from offender-centric to victim-centric justice
28	2023, S. K. Nigam et al.	Legal Question-Answering in the Indian Context: Efficacy and Challenges	Comparative exploration of AI frameworks using OpenAI GPT	Indian criminal legal landscape datasets	High proficiency in decoding natural language prompts
29	2024, N. Shah et al.	On the Analysis of a BERT-based Domain-Specific Question Answering Models	Analysis of BERT-based extractive QA models	Indian Constitutional and statutory datasets	Demonstrates efficacy in decoding natural language legal prompts
30	2025, V. Volodymyr et al.	AI and Cybercrime: New Challenges and Prospects for Regulation	Review of AI's role in cybercrime and legislative needs	International and regional cybercrime regulations	Identified critical gaps in laws regarding AI-generated crimes
31	2026, Dr. C. K. Yadav	Overhauling India's Criminal Law Framework: A Doctrinal Comparison of BNS 2023 with IPC 1860	Doctrinal and comparative methodologies	BNS 2023, IPC 1860, BNSS, and BSA	Analyzed key structural changes like offense reduction and new specific crimes
V.	**Comparative & Specialized Legal Studies**				
32	2025, F. Ariai et al.	Natural Language Processing for the Legal Domain: A Survey	Systematic survey of LLM applications, RAG, and fine-tuning	Global legal NLP datasets and benchmarks	Identified key trends in judgment

					prediction and contract analysis
33	2025, M. Hindi et al.	Enhancing Precision and Interpretability of RAG in Legal Technology: A Survey	Survey of advanced RAG: query expansion, re-ranking, and self-correction	Global survey of RAG implementations in legal tech	Detailed roadmap for building robust, interpretable legal AI
34	2025, A. Z. Pratama et al.	RAG for Indonesian Criminal Law Information Using the LLaMA Model	RAG system utilizing LLaMA-3-8B-Instruct with ChromaDB	Indonesian Criminal Code (KUHP) documents	High accuracy and faithfulness in generating law info
35	2025, S. Amri et al.	Moroccan legal assistant Enhanced by Retrieval-Augmented Generation	RAG-based assistant using BERT embeddings and GPT for responses	Moroccan legal codes and labor law documents	Reduced hallucinations and improved accuracy in legal advice
36	2020, E. Leitner et al.	A Dataset of German Legal Documents for Named Entity Recognition	Manual annotation of 19 fine-grained semantic classes	67,000 sentences from German federal court decisions	Provided a high-quality gold standard for German legal NER
37	2025, M. Sameer et al.	Cybercrime under the new Iraqi draft cybercrime law	Jurisdictional analysis of modern cybercrime drafting and digital evidence	Iraqi draft cybercrime laws and international standards	Provided a cross-border perspective on digital evidence legalities
38	2025, Anonymous	CASEGPT: A CASE REASONING FRAMEWORK	Synergizes LLMs and RAG technology for contextual understanding	Medical and legal field datasets (case data)	12% improvement in legal precedent retrieval precision
39	2025, S. Ni et al.	Three-Stage Framework for Legal Question Answering	Pre-training, fine-tuning, and re-ranking framework (PFR-LQA)	Large-scale database of legal question-answer pairs	Boosted dense retrieval performance and text representation

Chapter 4: INFERENCE FROM LITERATURE REVIEW

The analysis of the 39 collected research papers yields four critical inferences:

1. ***Architecture:***
   Retrieval-Augmented Generation (RAG) is the only reliable method for legal AI in 2026, as it grounds LLM outputs in verified statutory text (BNS/IT Act), effectively eliminating "hallucinations" of non-existent laws.

2. ***Linguistic Complexity:***
   Research shows that "Matrix Language Identification" is essential for Hinglish. Standard English models miss the semantic intent of code-mixed queries, requiring specialized fine-tuning on datasets like COMI-LINGUA.

3. ***The BNS Shift:***
   The transition from IPC to BNS is not a simple 1:1 mapping. Specific sections regarding organized crime (Sec 111) and digital evidence (BSA protocols) require the system to have a dynamic, updated knowledge base.

4. ***Actionable Forensics:***
   Literature indicates that a guidance system is only effective if it performs "Named Entity Recognition" to extract evidence (transaction IDs) that can be immediately used in a FIR or bank freeze request.

Chapter 5: OBJECTIVES

- **To Develop a Robust Hinglish NLP Pipeline:**
  Build a system capable of accurately identifying legal intent from informal, code-mixed Indian dialects in both Roman and Devanagari scripts.
- **To Implement an Accurate BNS-IT Act Mapping Engine:**
  Create a localized vector database that maps extracted crime entities to the correct legal sections of the 2024 Bharatiya Nyaya Sanhita.
- **To Automate Procedural Handout Generation:**
  Design a module that converts legal advice into actionable "handouts" and pre-filled PDF complaint forms for the National Cyber Crime Portal.
- **To Provide Precedent-Based Context:**
  Integrate a retrieval module that fetches similar court cases and police records to show users how specific laws have been applied in past fraud instances.

Chapter 6: ANALYSIS

The current state of cybercrime reporting in India suffers from high friction. When a user is defrauded, they typically search Google for "what to do," leading to a sea of generic, often outdated information. There is no automated bridge that takes a user's specific story (e.g., "Mera UPI pin leak ho gaya and balance zero ho gaya") and tells them exactly which law applies.

The analysis shows that a "Victim-Centric" model is required. By analyzing the linguistic patterns of victims, we find that the emotional state during the "Golden Hour" prevents them from filling out complex legal forms. The proposed system solves this by acting as an intermediary that does the "heavy lifting" of legal drafting. Technically, the system must overcome the high false-positive rates seen in general NER models when dealing with legal terminology. The use of a "Semantic Similarity Filter" alongside Legal-BERT is necessary to ensure that "fraud" is not confused with civil "disputes" in the system's reasoning.

Chapter 7: PROPOSED SYSTEM

The **Cyber Legal Guidance System** is built on a four-tier architecture:

1. **User Interaction Tier:**
   A web/mobile interface where users input queries in Hinglish. This tier also supports image uploads for OCR processing of fraud-related screenshots (Bank SMS, WhatsApp chats).

2. **NLP & Translation Tier:**
   Utilizes a fine-tuned transformer model to identify the "Matrix Language" and extract entities. It translates the core grievance into a formal legal problem statement.

3. **Techno-Legal Reasoning Tier:**
   A RAG engine compares the problem statement against the BNS 2023 and IT Act 2000. It uses a Knowledge Graph to identify "Similar Cases" based on the extracted entities (e.g., phishing, vishing, or identity theft).

4. **Handout Generation Tier:**
   The final output is an "Information Handout." This includes:
   (a) Applicable BNS sections,
   (b) Step-by-step recovery instructions, and
   (c) A pre-filled complaint form containing the extracted forensic evidence.

Example:

Raw Input :

“Mera phone hack ho gya aur bank se ₹50000 cut gaye. TXN ID 987654321 hai.”

Tokenization and Lang ID :

“Mera phone hack ho gya aur bank se ₹50000 cut gaye. TXN ID 987654321 hai.”

Legal – Bert Entity Extraction:

Amount : ₹50000
Txn_ID: 987654321
Tags: {hack: financial theft}

Translation & Vector Search

Query: “Financial theft via compromised device”

Final Legal Citations

IT Act Sec 66 (Computer Related Offences)
BNS Sec 318 (Cheating)
Steps to take: Call 1930 helpline number and then call your bank to report the unauthorized transaction.
Confidence Score : 99 %

Chapter 8: CONCLUSION

The proposed Cyber Legal Guidance System addresses the critical limitations observed in current legal-tech and NLP research. By focusing specifically on the Indian sociolinguistic context and the updated BNS framework, the system removes the technical and linguistic barriers that prevent victims from seeking justice. Decisive action in cybercrime is time-sensitive; by providing immediate, accurate, and actionable legal handouts, this system ensures that the legal "Golden Hour" is utilized to its fullest, moving the Indian criminal justice system closer to a truly victim-centric model.

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